WEEK 01

Values, types and execution

តម្លៃប្រភេទ និងការប្រតិត្ត

This week: learn, practise, build

សប្តាហ៍នេះ រៀន អនុវត្តនិងបង្កើត

01 · Python

Values, types and execution
Understand the idea and trace examples.

02 · Practice

Practise the same Python concept.
Predict, change, diagnose and explain.

03 · Project

Labelled planning calculation
Build, discuss and test with a peer.

One weekly component moves the final project forward.

ផ្នែកមួយរាល់សប្តាហ៍ជួយឱ្យគម្រោងចុងក្រោយរីកចម្រើន។

Python foundations

មូលដ្ឋាន Python

Values, types and execution
Concept → trace a small example → explain its behavior

បែងចែកឈ្មោះ និងតម្លៃហើយតាមដានលំដាប់ប្រតិត្តិ

Names, values and statements

ឈ្មោះ តម្លៃនិងសេចក្តីថ្លែង

A value has a type

Integers support arithmetic; strings store text. Quoted text is a literal.

Assignment binds a name

The right side runs first. A later assignment changes what the name refers to.

Execution has an order

Run statements from top to bottom. Use print to observe a value at that point.

Understand the behavior before changing the example.

យល់ឥរិយាបថក្នុងមុនកែឧទាហរណ៍។

Python executes statements in order

Python ប្រតិបត្តិសេចក្តីថ្លែងការណ៍តាមលំដាប់

1

Assign

count = 4
unit_size = 3
Names refer to values.

2

Evaluate

total = unit_size * count
Read values; multiply.

3

Use the result

print(total)
Display the stored value.

Assignment stores a value. Changing an input does not rerun an earlier expression.

ការកំណត់តម្លៃក្បាលម៉ូល។ ប្តូរធាតុចូលមិនគណនាកន្សោមមុនឡើងវិញទេ។

Syntax to read and reuse

វាក្យសម្ពន្ធសម្រាប់អាន និងប្រើឡើងវិញ

Python / syntax	Meaning	When to use it
name = value	Assign a value	Keep an input for later
3 * 4	Multiply numbers	Calculate a quantity
print("Total:", total)	Display a value	Label output and units
# explanation	Comment; not executed	Explain an assumption

Python Basics · sections 00, 01, 02, 03, 14 · use only when you need a lookup.

Python Basics ផ្នែក 00, 01, 02, 03, 14៖ មើលយោងតែពេលចាំបាច់។

Names and text are different

ឈ្មោះអថេរ និងអត្ថទខុសគ្នា

```
count = 4
unit = 3
total = count * unit
print(count)
print("count")
print(type(count).__name__)
print(total)
```

1. A name retrieves its current value.
2. Quotes make text, not a variable lookup.
3. Expected: 4, count, int, 12.

Predict → execute → explain the value at each step.

ព្យាករ → ដំណើរការ → ពន្យល់តម្លៃនៅជំហានមួយៗ។

Variables and string literals

អថេរ និងអត្ថន័យនៃសញ្ញាស្រម៉ង់

```
containers = 4
print(containers)
print("containers")
print(type(containers).__name__)
```

1. Predict all three outputs before running
2. Assignment associates a name with a value
3. Change the value and predict again

Quotes create text; an unquoted name retrieves its assigned value.

សញ្ញាស្រម៉ង់បង្កើតអត្ថន័យ; ឈ្មោះគ្មានសម្រង់ទាញតម្លៃថេរ។

Numbers and strings behave differently

លេខ និងអត្ថបទមានឥរិយាបថខុសគ្នា

```
print(3 * 4)
print("3" * 4)
assert 3 * 4 == 12
```

1. How does quoting the first value change the output?
2. Container counts use integer values
3. Keep expectations independent of output

Valid code can still express the wrong operation.

កូដត្រឹមវាក្យសម្ពន្ធអាចនៅតែធ្វើការងារខុស។

Assign, calculate and print

កំណត់តម្លៃគណនា និងបង្ហាញ

```
description = "Practice meal containers"
portions_per_container = 3
containers = 4
planned = portions_per_container * containers
print(description)
print("Containers:", containers)
print("Planned portions:", planned)
assert planned == 12
```

1. Assign input values first.
2. Use an expression to calculate a result.
3. print() displays the stored result.

Keep inputs, calculation and display as separate statements.

បំបែកការកំណត់ទុក ការគណនា និងការបង្ហាញ។

Execution order: when is a value recalculated?

លំដាប់ប្រតិបត្តិពេលណាគណនាតម្លៃឡើងវិញ?

```
portions_per_container = 3
containers = 4
planned = portions_per_container * containers
containers = 2
print("Before:", planned)
planned = portions_per_container * containers
print("After:", planned)
assert planned == 6
```

1. Predict the value printed before recalculation
2. Then predict the value printed after recalculation
3. Rerun the calculation after changing inputs

Changing an input does not rerun an earlier multiplication.

ការផ្លាស់ប្តូរធាតុចូលមិនធ្វើឲ្យមានការគុណមុនៗឡើងវិញទេ។

One language, two execution environments

ភាសាតែមួយ បរិស្ថានប្រតិបត្តិ

Notebook: .ipynb

Narrative and code cells share a persistent kernel. Run cells and inspect their results.

Source file: .py

A new interpreter reads saved statements from top to bottom. Its output appears in a terminal.

A new script process does not inherit notebook variables.

ដំណើរការ scriptថ្មីទទួលអថេរពីnotebook ទេ។

Save and run a Python file

រក្សាទុក និងដំណើរការឯកសារ Python

Context	Command or observation
Working directory	learning_materials/
Terminal	python programs/first_pickup.py
Choose Python	Use your course interpreter; see programs/README.md
Default calculation	3 portions/container × 4 containers
Expected meaning	12 planned portions, not delivered impact

Replace python with your course interpreter; save before rerunning.

ប្រើinterpreter របស់វគ្គសិក្សា python; រក្សាទុកមុនរត់ឡើងវិញ។

Hands-on practice

លំហាត់អនុវត្តផ្ទាល់

Week_01.ipynb · Session 2
Predict → run → change → explain

ព្យាករ រត់ ប្តូរ និងពន្យល់លទ្ធផល។

Your turn: practise the same Python skills

ដល់វេនអ្នក៖ អនុវត្តជំនាញ Python ដដែល

Change one value

Use 3 portions/container.
Try 4, then 2 containers.
Predict before running.

Trace execution

Change only the description.
Then change the count.
Which lines need rerunning?

Run a saved file

Copy first_pickup.py.
Set the count to 3.
Save, run, compare.

Use the matching exercises in this weekly notebook. Explain what changed and why.

ប្រើលំហាត់ត្រូវគ្នានៅក្នុង notebook សប្តាហ៍នេះ។ ពន្យល់អ្វីដែលបានផ្លាស់ប្តូរ និងមូលហេតុ។

Check your work before asking for help

ធ្វើការងារមុនសុំជំនួយ

Predict first

Write what you expect before executing.

Change one thing

Compare values, types, paths or returned results.

Explain the cause

Use the relevant Python rule. Retest your repair.

Explain names versus values, number/string operations and execution order.

ពន្យល់ឈ្មោះធៀបតម្លៃប្រតិបត្តិការលេខនិងអត្ថបទ និងលំដាប់វគ្គ។

Project studio

វគ្គសិក្សារចនាគម្រោង

Labelled planning calculation
Short design briefing → most time for your work

គណនាចំណែកតាមផែនការ និងបង្ហាញឯកតាឱ្យច្បាស់។

The people’s task this week

កិច្ចការរបស់អ្នកច្រើនាសប្តាហ៍នេះ

1

Someone needs help

The coordinator asks: how much are we planning?

2

This week, you will

Use the declared counts to calculate and label planned portions.

3

A useful result

The coordinator can read the estimate and ask about unclear units.

Use the supplied starting files. Focus on one useful result and check it.

ប្រើកសារចាប់ផ្តើមដែលផ្តល់។ ផ្តោតលើលទ្ធផលមានប្រយោជន៍មួយ ហើយធ្វើតេស្ត។

Trace a declaration into a planning note

តាមដានឧស្ម័យដែលបានប្រកាសទៅកំណត់ត្រាផែនការ

1

Declared input

3 planned portions in each specified container

2

Proposed count

4 containers in this practice plan

3

Calculation

3 multiplied by 4 gives 12 planned portions

4

Labelled output

Show the units and retain planning status

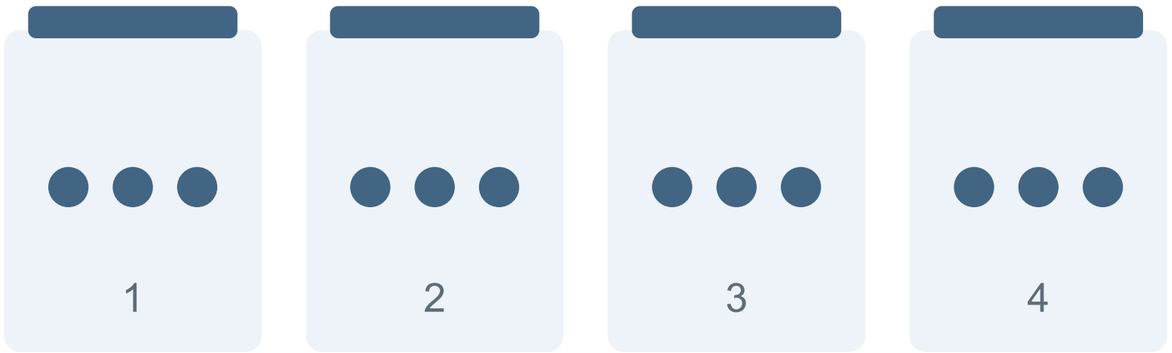
Portions per container multiplied by containers gives planned portions.

ចំណែកដង្ហមួយប្រអប់ × ចំនួនប្រអប់ = ចំណែកតាមផែនការ។

See the values behind the calculation

ធ្វើឲ្យតម្លៃដលប្រើការគណនា

4 containers



3 declared portions per container

3 × 4 = 12

Planned portions

A calculation from declared inputs.

Read the diagram, then trace the same steps in code.

អានដ្យាក្រាម រួចតាមដានជំហានដដែលក្នុងកូដ។

Your implementation TODO

កិច្ចការដែលអ្នកត្រូវសរសេរក្ដរ

Supplied starting point

A notebook starter provides variable names and print calls.

You implement / do

Set description, portions_per_container=2 and containers. Calculate and display planned_portions.

Check the evidence

Run containers 1, 3 and 0: expect 2, 6 and 0. Explain why this is a plan, not recorded delivery.

Explain the input units, the calculation and one coordinator question.

ពន្យល់ឯកតាទិន្នន័យ ការគណនា និងសំណួរមួយសម្រាប់អ្នកសម្របសម្រួល។

Show the component working

បង្ហាញថាផ្នែកនេះដំណើរការ

Step	What to show
1 · Locate	Identify the input, this component and its next user.
2 · Implement	Print description, input counts and planned portions.
3 · Check	2 portions/container × 1, 3, 0 containers → 2, 6, 0.
4 · Explain	Show your code, its actual result and why it behaves that way.

Explain the input units, the calculation and one coordinator question.

ពន្យល់ឯកតាធ្វើយ ការគណនា និងសំណួរមួយសម្រាប់អ្នកសម្របសម្រួល។

Further reading in English

ឯកសារអានបន្ថែមជាភាសាអង់គ្លេស

1 Python tutorial: numbers and strings

https://docs.python.org/3/tutorial/introduction.html

2 Python tutorial: input and output

https://docs.python.org/3/tutorial/inputoutput.html

Read one relevant section, then bring an example and a question.

អានផ្នែកដែលពាក់ព័ន្ធមួយមកឧទាហរណ៍មួយ និងសំណួរមួយមកវិភាគក្សា។